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PRG 400
Week 1 Assignment

Functional programming is the approach to programming, in which we manipulate data using functions. We can replace long loops with in-built functions such as, map, filter and reduce as well as with lambda functions to write shorter and cleaner codes.

Lambda Function

A lambda function is a small anonymous function (a nameless function). It is typically applied on small tasks where one does not need to write a complete function.

```
In [1]: # Normal function
def square(x):
    return x * x

# Lambda function
square_lambda = lambda x: x * x

# Using lambda
print(square_lambda(5))
```

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Code Explanation

- First, a normal function square is defined.
- After that a lambda is written to accomplish the same task in a single line.
- lambda x: x * x: take input x and output x * x.

Program Flow

1. An anonymous function is defined.
2. The value of 5 is taken to the function.
3. The function returns 25 .

Map Function

The `map()` method is used to perform a function on each element of a list. It is applied when we wish to convert everything in a sequence.

```
In [3]: numbers = [1, 2, 3, 4, 5]

# Using lambda with map
squared_numbers = list(map(lambda x: x * x, numbers))

print(squared_numbers)
```

```
[1, 4, 9, 16, 25]
```

Code Explanation

- A list of numbers is developed.
- `map(s)` is used to run the lambda function over every item.
- Each number is squared.

Program Flow

1. The list `[1,2,3,4,5]` is created.
2. The use of the lambda function on every number is implemented by using `map`.
3. The squared values are returned as a new list.

Filter Function

The `filter()` method is used to pick desired elements out of a list as a result of a condition.

```
In [4]: numbers = [1, 2, 3, 4, 5, 6]

# Get even numbers
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))

print(even_numbers)
```

```
[2, 4, 6]
```

Code Explanation

- The list of numbers is created.
- `lambda` is used to test whether a number is even.

- filter removes all the numbers that do not match the condition.

Program Flow

1. The list `[1,2,3,4,5,6]` is created.
2. The condition is the checking of each number with the aid of the condition $x \% 2 = 0$.
3. Even numbers are only returned.

Reduce Function

The function reduce is applied in order to combine the elements of a list into one value. It performs a function several times on the elements.

```
In [5]: from functools import reduce

# Reduce Example

numbers = [1, 2, 3, 4, 5]

# Sum of all numbers
total = reduce(lambda x, y: x + y, numbers)

print(total)
```

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Code Explanation

- The importation of the reduce function uses functools.
- The lambda function sums two numbers simultaneously.
- It repeats the values combining process until only one result is left.

Program Flow

1. Begin with the first two numbers $1 + 2 = 3$.
2. Then $3 + 3 = 6$
3. Then $6 + 4 = 10$
4. Then $10 + 5 = 15$
5. Final result is returned.